

The Price of the Early Exit: ncu After GPU Periodicity

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Binary: `binary-v7-gpu-periodicity` (2abc909, on main since 41b6c2e), captured with Nsight Compute 2026.2. Baseline: experiment 16’s tracked capture of `binary-v5-affinity` (fc33e29) — the device `diverge()` is unchanged v5→v6, so that capture is the immediately-pre-check kernel. Raw data: `experiments/24-ncu-gpu-periodicity/`.

Resumen

Reports 16/17 pinned this kernel at exactly 100.0% warp execution efficiency on interior content, and report 21 cited that number to defer the GPU-side periodicity check (“expect a loss”). Report 23 falsified the conclusion at the wall (−20.5% hybrid); this report replaces the stale counters. Same method as experiment 16 (same four content-regime specs, same kernel caps, same metrics; aggregation validated to reproduce experiment 16’s summary exactly; clocks match within 1%, so durations compare). The price of the early exit is small and precisely locatable: interior warp efficiency falls only **100.0** → **96.3**% (minimum kernel 95.0%) while interior kernels get **9.9**× faster; the isolated overhead — pure-exterior content, where the check never fires — is +**12.4**%, half the 25–30% estimate, because `nvcc` short-circuits the two anchor compares into one always-executed DSETP. The cost concentrates on boundary content (+23%, kernels production routes to the CPU anyway); the production mix nets 9.4× per kernel. The FP64 pipe stays at ~85% of peak in every regime, before and after: the kernel remains FP64-bound, and every metric that “degraded” — warp efficiency, achieved occupancy — reads worse precisely because the work now ends early.

1. Motivation: replace the number the wrong prediction leaned on

Report 16 measured interior kernels at exactly 100.0% warp execution efficiency and 85.6% of FP64 peak; report 17 reproduced both on the four zoom-point geometries. Those were correct measurements read as health: report 21’s recommendation 6 cited them to predict the GPU-side periodicity check would lose. Report 23 measured the check instead — mix-controlled kernel 2.34×, hybrid 29.83 → 23.72 s, output byte-identical — and identified the misreading: on interior content, “100% efficiency, zero early exits” *is* the waste (every lane of every warp grinding provably-useless iterations to MAXITER in perfect lockstep). What report 23 could not say is what the counters look like now: how much warp efficiency the early exit actually spends, where the two-compares-per-iteration overhead lands, and whether the FP64 roofline moved. Without a measured replacement, the 100.0% baseline stays on record as a target someone could “restore” by deleting the check. This experiment re-baselines all of it.

2. Methodology

Identical to experiment 16, on the merged v7 binary: the same four content-regime specs reused verbatim (*interior* = main cardioid, **maxIter** 2000–3000; *exterior* = fast escape; *boundary* = seahorse valley; *canonical* = the production zoom), GPU-only runs (1 1 0.1 **32768** 1 0 — the single GPU thread sees every region class), **—set full** on 6/6/60/60 kernels plus a cheap warp/branch **—metrics** pass on 6/6/300/300. Counters are admin-gated; capture ran via **sudo capture.sh**, analysis unprivileged via **analyze.py**.

Baseline. Experiment 16’s tracked **analysis/summary.csv**. The device **diverge()** is unchanged between **binary-v5-affinity** (which experiment 16 captured) and v6 (a CPU-only change), so no re-capture of the old kernel is needed. **analyze.py** reproduces experiment 16’s aggregation exactly — validated in-session against its tracked summary on all four regimes (simple mean over captured kernels of the same raw-page columns). One slip found in the reference: experiment 16’s *exterior dur_ms* is recorded in μs (614.869); corrected to 0.615 ms here.

Comparability. Warp/branch efficiency, pipe and SOL percentages, and occupancy are clock-independent ratios. Durations normally carry a thermal caveat across sessions, but the recorded GPC clocks match within 1% (v6: 1.317–1.345 GHz; v7: 1.322–1.342 GHz), so the duration ratios are quoted as measured. The decomposition is deterministic and identical in both binaries (report 23), so capture N sees the same N regions.

3. Results

Tabla 1: Full-set captures, v6 baseline (experiment 16) vs v7 (this capture); each cell is v6 $\rightarrow$ v7. Read row-wise: the check’s entire metric cost is the interior warp-efficiency drop, the +12–23% on content that never cycles, and 6 registers; the FP64 pipe does not move.

metric	interior	exterior	boundary	canonical
warp exec. eff. (%)	100.0 $\rightarrow$ 96.3	100.0 $\rightarrow$ 100.0	92.6 $\rightarrow$ 92.4	94.4 $\rightarrow$ 91.9
branch eff. (%)	100.0 $\rightarrow$ 99.7	100.0 $\rightarrow$ 100.0	99.6 $\rightarrow$ 99.8	96.4 $\rightarrow$ 97.8
FP64 pipe (% peak)	85.6 $\rightarrow$ 85.3	85.6 $\rightarrow$ 84.9	85.3 $\rightarrow$ 85.1	84.1 $\rightarrow$ 84.0
compute SOL (%)	85.5 $\rightarrow$ 85.1	85.4 $\rightarrow$ 84.6	78.1 $\rightarrow$ 77.9	69.4 $\rightarrow$ 66.6
achieved occup. (%)	98.9 $\rightarrow$ 89.5	94.0 $\rightarrow$ 91.5	84.3 $\rightarrow$ 84.6	79.1 $\rightarrow$ 77.4
mean kernel (ms)	726.4 $\rightarrow$ 73.5	0.615 $\rightarrow$ 0.691	16.8 $\rightarrow$ 20.7	30.7 $\rightarrow$ 3.25
duration ratio	9.9 $\times$	+12.4%	+23.3%	9.4 $\times$
registers/thread	22 $\rightarrow$ 28	22 $\rightarrow$ 28	22 $\rightarrow$ 28	22 $\rightarrow$ 28

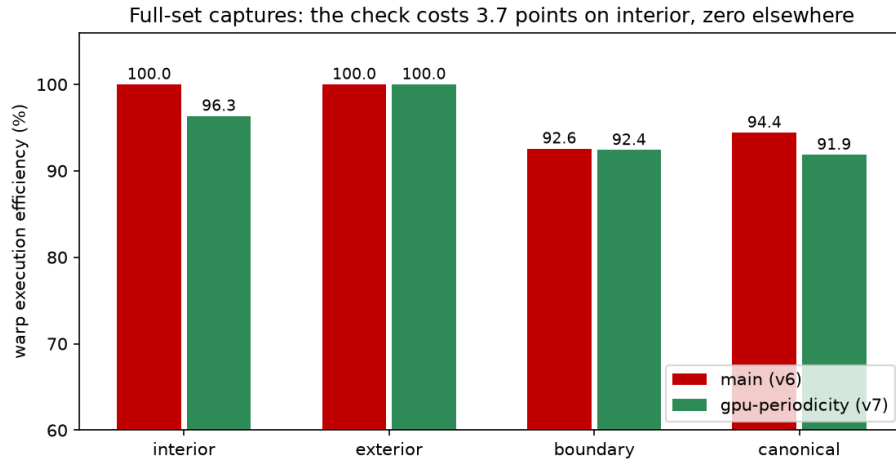


Figure 1: Full-set warp execution efficiency by regime. The check’s entire efficiency cost on the full-set captures is the 3.7-point interior drop (and 2.6 on the production mix); exterior — where no orbit ever cycles — is untouched at 100.0%, and boundary’s structural floor-leaf divergence (report 16) is unchanged.

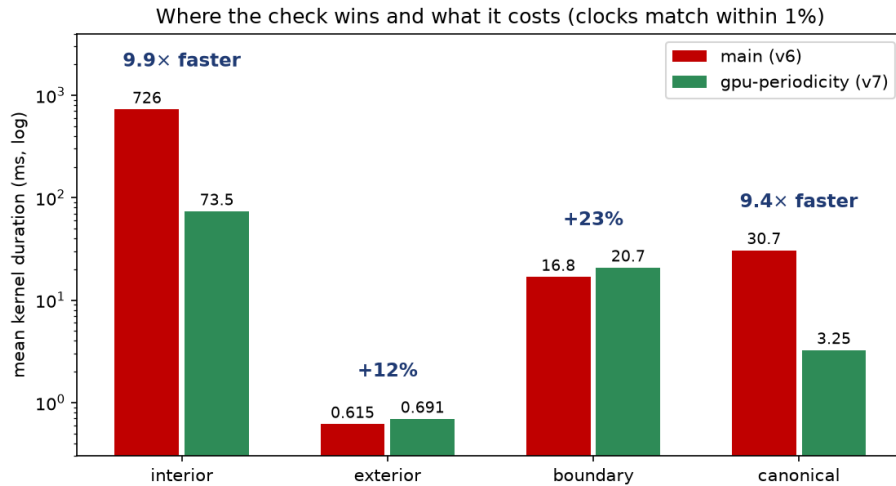


Figure 2: Mean kernel duration per regime, log scale (GPC clocks match within 1% between the sessions). The win lands where the cycles are (interior 9.9×, production mix 9.4×); the cost lands where they aren’t (exterior +12.4%, boundary +23.3%). Production affinity routes boundary floor leaves to the CPU pool, so the +23% barely prices into the production config that report 23 measured at −20.5% wall.

Tabla 2: Per-kernel warp-efficiency distribution from the large `_warp` captures (both binaries reduced identically). Boundary is statistically unchanged — the floor-leaf divergence is a property of region size, not of the check — while canonical’s distribution shifts down: its big interior kernels now contain the early exits, by design.

regime	kernels	mean (%)	min (%)	below 70%
boundary v6	276	70.7	28.6	47.1%
boundary v7	276	70.2	28.5	47.5%
canonical v6	300	87.6	22.2	12.7%
canonical v7	300	78.1	18.7	32.0%

Divergence structure: boundary unchanged; canonical’s interior kernels absorb the early exits

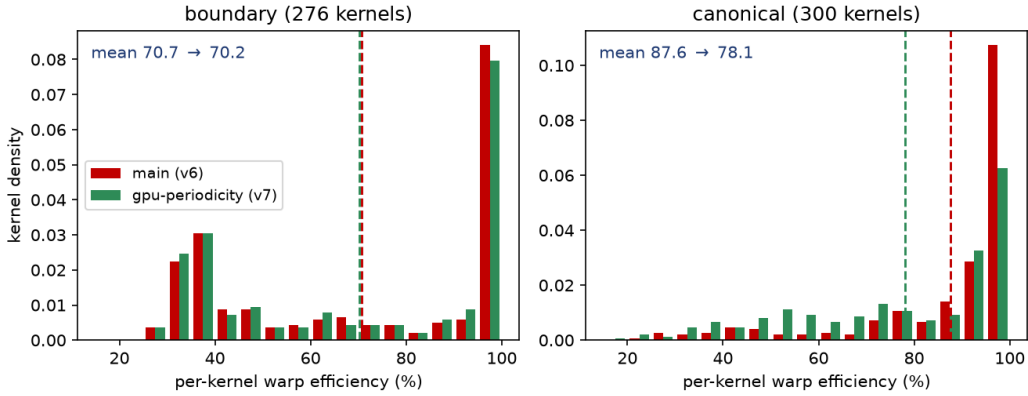


Figure 3: The same distributions as Table 2. Left: boundary histograms lie on top of each other — the check neither adds to nor removes the structural divergence of the 240×135 floor leaves. Right: canonical’s mass moves out of the 95–100% bin into the 40–90% range: the interior kernels that used to run in wasteful lockstep now retire lanes as they detect their cycles.

4. Analysis

4.1. Interior pays 3.7 points because detection is synchronized within a warp

The feared divergence catastrophe is a 3.7-point drop: interior kernels average 96.3% warp efficiency (minimum 95.0%, nothing below 90% in any capture) while running $9.9\times$ faster. The reason detection is so coherent: a warp’s 32 lanes are contiguous pixels of the same interior region converging to the same attracting cycle, and every lane refreshes its Brent anchor at the *same* iteration numbers (8, 16, 32, ... — the schedule is iteration-indexed, not state-indexed). Neighbouring orbits therefore satisfy the exact-revisit test at nearly the same iteration, and the masked-idle tail between the first and last detection in a warp is short relative to even the shortened kernel. The duration ratio quantifies the effective latency: $9.9\times$ at `maxIter` 2000–3000

and +12.4% per-iteration overhead implies the slowest lane detects at ~ 225 iterations on main-cardioid content — between report 21’s 67 (period-3 bulb) and ~ 900 (deep minibrot), as expected for the period-1 cardioid interior with a slow transient.

4.2. The isolated overhead is +12.4%, not 25–30%: nvcc short-circuits the compare pair

Exterior content prices the check’s overhead with zero benefit mixed in: no escaping orbit ever revisits a state, so the branch never fires, and the regime stays at exactly 100.0% warp efficiency — the +12.4% duration is pure per-iteration cost. Reports 21/23 estimated 25–30% by counting two FP64 compares against a ~ 7 -instruction recurrence; the SASS shows why the measured cost is half that: the compiler compiles `vx == sx && vy == sy` as one always-executed `DSETP.NEU` on `vx`, with the `vy` compare (`DSETP.EQ`) predicated behind it — it almost never issues on non-cycling orbits. One extra FP64-pipe instruction on a loop body of ~ 8 predicts +12.5%; measured, +12.4%. Consistently, the FP64 pipe stays at $\sim 85\%$ of peak in every regime: the check did not move the roofline, it added one slot to the iteration.

4.3. Boundary pays the most (+23%) and produces nothing — and production doesn’t care

Boundary kernels slow $16.8 \rightarrow 20.7$ ms with a warp-efficiency distribution that is statistically identical (mean $70.7 \rightarrow 70.2$, 47% below 70% in both, minimum $\sim 28.5\%$): seahorse-valley content iterates long but either escapes or fails to *exactly* cycle within the anchor schedule, so it pays the compare on every iteration and the early exit almost never fires. The +23% exceeding the +12.5% static floor is the one number these counters do not fully attribute (candidates: the predicated second compare firing on near-cyclic orbits, and thinner warps hiding the recurrence latency less well); it is also the least consequential — report 16 showed these are the 240×135 pixel-floor leaves, and in the production hybrid the affinity queue routes them to the CPU pool. The production mix (canonical) nets $9.4\times$ per kernel regardless.

4.4. The occupancy drop is the early exit’s signature, not register pressure

Registers rise $22 \rightarrow 28$ (the two anchor doubles and the schedule counter live across the loop), which moves the register occupancy limit from 10 to 8 blocks/SM — both far above the 4-block warp limit that actually binds ($4 \times 256 = 1024$ threads/SM), so *theoretical* occupancy is unchanged. The achieved drop ($98.9 \rightarrow 89.5\%$ on interior) is the same phenomenon as the warp-efficiency drop seen one level up: warps retire as their lanes detect cycles, blocks drain, and the SM spends a larger fraction of a much shorter kernel partially full. Like warp efficiency, achieved occupancy now reads “worse” *because* the kernel stops doing useless work.

4.5. The counters reconcile with report 23’s $2.34\times$, and the metrics’ role changes

Report 23 measured the mix-controlled kernel total at $2.34\times$ over all 5,608 regions; these captures show interior-type content at $\sim 9.9\times$ and never-cycling content at

$\sim 1.2\times$ slower. The blend $s/9.9 + (1 - s) \times 1.2 = 1/2.34$ solves to $s \approx 0.70$: seventy percent of the v6 kernel time was interior grind — the GPU-side echo of the leaf-purity census that motivated report 21. The practical conclusion for the metric suite: on this kernel, warp execution efficiency and achieved occupancy are no longer health indicators; they are the accounting of eliminated work. The FP64 pipe percentage ($\sim 85\%$, invariant) remains the real health line — it says the kernel is still FP64-bound, so the next kernel-side lever (if one is ever needed) is still arithmetic, not scheduling.

5. Recommendations

1. **Adopt these numbers as the healthy baselines:** interior 96.3% warp / 99.7% branch efficiency, canonical distribution mean 78.1% with 32% of kernels below 70%, achieved occupancy 89.5%, FP64 pipe 85%. Reports 16/17’s 100.0% figures are superseded by design; a future capture that reads 100.0% on interior content means the check was removed and $\sim 10\times$ kernel time came back.
2. **Do not pursue k -strided compare amortization.** The motivation (reports 21/23’s 25–30% overhead estimate) is gone: the short-circuited compare costs +12.4%, and striding the anchor compare to every k -th iteration would delay detection by up to $k-1$ iterations on a ~ 225 -iteration prize to recover part of one instruction in eight. The measured ceiling is not worth the exactness-argument complication.
3. **Proceed to the scanLine commit path** (report 23, recommendation 2). The kernel side is done: $9.4\times$ on the production mix with the pipe still 85% saturated leaves the GPU thread’s host side (12.8 s vs 10.7 s kernel at steady state) as the only lane cost that has not been attacked.
4. **Port the check to oclkernel.cpp before the third iGPU repricing** (report 23, recommendation 4) — unchanged, with one addition from this data: expect the iGPU’s kernel-side gain to be content-identical ($\sim 10\times$ on interior) but its warp (subgroup) metrics to shift the same way; budget the same identity-gate-against-itself validation.

6. Conclusion

The GPU-side periodicity check’s microarchitectural price is measured and small: interior warp execution efficiency $100.0 \rightarrow 96.3\%$ (minimum 95.0%), achieved occupancy $98.9 \rightarrow 89.5\%$, six extra registers with the theoretical occupancy limit untouched, and +12.4% per-iteration overhead where no cycle exists — half the pre-measurement estimate, because `nvcc` short-circuits the anchor compares. The prize the metrics paid for: interior kernels $9.9\times$, the production mix $9.4\times$ per kernel, reconciling to report 23’s mix-controlled $2.34\times$ via a 70/30 interior/other time split. Boundary floor leaves pay +23% and remain exactly as divergent as report 16 found them — structural, size-bound, and routed to the CPU in production. The FP64 pipe stays at $\sim 85\%$ of peak in every regime: the kernel is still FP64-bound, the roofline did not move, and the metrics that “got worse” are the ledger of the work that no

longer happens. Reports 16/17's 100.0% baselines are hereby replaced; the queued levers are unchanged: **scanLine** host-side commit, then the iGPU port and repricing.